

EXPERIMENT - 2

Title: Taxonomy of Recommender Systems

Aim: The aim of this experiment is to study the taxonomy of recommender systems and understand how different recommendation approaches are classified based on personalization, data usage, and recommendation strategy.

Objectives: The objective of this experiment is to understand the need for classifying recommender systems, learn the difference between personalized and non-personalized recommendations, study major types of recommender systems, and understand how these approaches are used in real-world applications.

Theory: In modern digital platforms, recommender systems play a crucial role in helping users discover relevant information from a very large set of available items. Since user preferences, data availability, and application requirements vary across platforms, a single recommendation approach is not sufficient for all situations. To handle this diversity, recommender systems are classified into different categories based on how they generate recommendations. This structured classification is known as the taxonomy of recommender systems and helps in understanding, designing, and implementing efficient recommendation solutions.

Taxonomy of Recommender Systems: The taxonomy of recommender systems refers to the systematic classification of recommendation approaches based on factors such as personalization level, data sources, and algorithmic techniques. This taxonomy helps developers and researchers understand how different recommender systems work and which approach is suitable for a particular application. Broadly, recommender systems are classified into non-personalized and personalized systems, with personalized systems further divided into content-based, collaborative filtering, and hybrid approaches.

Non-Personalized Recommender Systems: Non-personalized recommender systems provide the same recommendations to all users without considering individual preferences or behavior. These systems rely on general trends such as popularity, ratings, or sales statistics to generate recommendations. Since no user-specific data is required, non-personalized systems are simple to implement and do not suffer from cold start issues. However, the lack of personalization often results in lower user satisfaction, as recommendations may not match individual interests.

Personalized Recommender Systems: Personalized recommender systems generate customized recommendations by analyzing user preferences, behavior, and interaction history. These systems aim to provide relevant suggestions tailored to each user, leading to higher engagement and satisfaction. Personalized systems require sufficient user data and computational resources, but they are more effective than non-personalized systems in delivering meaningful recommendations.

Content-Based Recommender Systems: Content-based recommender systems recommend items that are similar to those previously liked or interacted with by the user. These systems analyze item features and build a user profile based on past behavior. Recommendations are generated by matching item characteristics with user interests. Content-based systems work well even when data from other users is limited, but they often suffer from limited diversity as users are repeatedly recommended similar items.

Collaborative Filtering Recommender Systems: Collaborative filtering recommender systems generate recommendations by analyzing similarities among users or items. The core idea is that users with similar

preferences in the past are likely to have similar interests in the future. These systems can recommend new and diverse items that a user has not previously explored. However, collaborative filtering systems face challenges such as cold start problems for new users or items and data sparsity when user interactions are limited.

Hybrid Recommender Systems: Hybrid recommender systems combine content-based and collaborative filtering approaches to overcome the limitations of individual methods. By integrating multiple recommendation strategies, hybrid systems improve accuracy, reduce cold start issues, and provide more diverse recommendations. Most modern recommender systems used in real-world applications adopt hybrid approaches to deliver better performance.

Applications of Recommender System Taxonomy: The taxonomy of recommender systems is widely applied across various domains such as e-commerce, entertainment platforms, educational systems, and social media. Different recommendation approaches are selected based on the nature of data and user requirements, making taxonomy an essential concept in recommender system design.

Result: Thus, the taxonomy of recommender systems was studied successfully. Various recommendation approaches including non-personalized, content-based, collaborative filtering, and hybrid recommender systems were understood along with their working principles and practical significance.